

Which experimental procedure(s) must scientists use to determine $V_{\max}$ and K_M of an enzymatic reaction using the Michaelis-Menten model? They must ensure that:

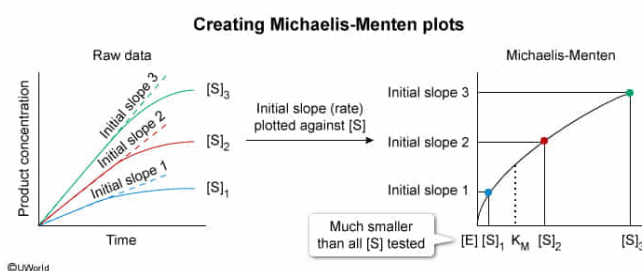
- I. they only measure the initial reaction rate for each substrate concentration.
- II. total enzyme concentration is much greater than the K_M of the reaction.
- III. each initial substrate concentration tested is much greater than enzyme concentration.

- ☐ A. I only [11%]
☐ B. III only [26%]
☐ C. I and II only [26%]
✔ ☒ D. I and III only [36%]

Correct

35% answered correctly

Explanation:



Michaelis-Menten kinetics are determined by measuring the rate of product formation in the presence of varying concentrations of substrate. The reaction rate (initial slope) at each substrate concentration is then plotted as a single point, with concentration on the x-axis and rate on the y-axis, yielding a hyperbolic curve (Michaelis-Menten plot) described by the [Michaelis-Menten equation](#).

The Michaelis-Menten equation relies on three assumptions:

- **The free ligand approximation** states that substrate concentration $[S]$ is constant during the reaction. This approximation is only true during the initial phase of the reaction, before a significant amount of substrate is

converted to product. Substrate can also be depleted when it **binds the enzyme** to form the enzyme-substrate complex (ES). To ensure that ES formation does not significantly impact [S], the total concentration of enzyme in solution should be much **smaller than any substrate concentration** tested (**Number III**).

- **The steady state assumption** states that the concentration of ES remains constant over the course of the reaction, allowing the rate of product formation to remain constant. Once [S] becomes **significantly depleted**, ES levels decrease and the reaction slows.
- **The irreversibility assumption** states that the reaction proceeds only in the forward direction, and product does not get converted back to substrate. Once enough **product accumulates**, the reverse reaction occurs at non-negligible levels and further slows the net rate of product formation.

Each of these assumptions holds true only during the **initial phase** of the reaction, before substrate is depleted or product accumulates. Therefore, for each substrate concentration measured, **only the initial reaction rate** should be reported (**Number I**).

(Number II) K_M is the substrate concentration at which $\frac{1}{2}V_{\max}$ occurs.

Because enzyme concentration must be substantially smaller than all substrate concentrations tested, enzyme concentration must be much **smaller than K_M** , not larger.

Educational objective:

The Michaelis-Menten equation models the initial rates of a reaction at various substrate concentrations. Rates are measured as the slope of the initial, linear phase of the reaction before substrate is depleted and product accumulates. Enzyme concentration should be much lower than all substrate concentrations tested.

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